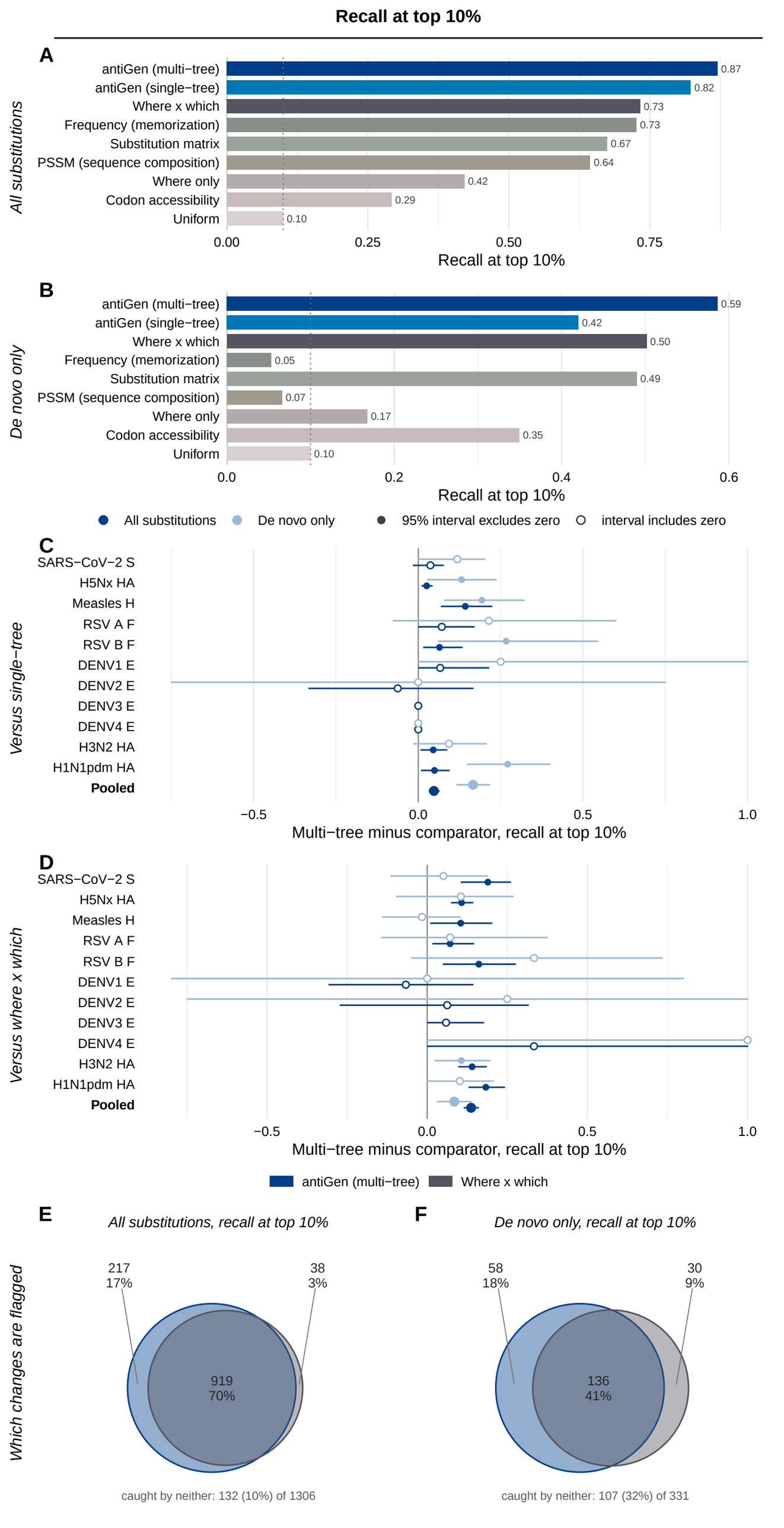
Recall at top 10%
A
All substitutions
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.87
0.82
0.73
0.73
0.67
0.64
0.42
0.29
0.10
0.00
0.25
0.50
0.75
Recall at top 10%
B
De novo only
0.59
0.42
0.50
0.05
0.49
0.07
0.17
0.35
0.10
0.0
0.2
0.4
0.6
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 10%
Which changes are flagged
217
17%
38
3%
919
70%
caught by neither: 132 (10%) of 1306
F
De novo only, recall at top 10%
58
18%
30
9%
136
41%
caught by neither: 107 (32%) of 331